

Banking Risk Analysis

This report outlines the technical workflow and key business insights derived from the banking dataset. Structured to demonstrate data wrangling, exploratory data analysis (EDA), and clear business intuition, this summary is formatted to be an excellent addition to a professional project repository.

1. Technical Environment & Data Preparation

The initial phase of the project focused on environment setup, data ingestion, and cleaning to ensure accurate downstream modeling.

- **Libraries Utilized:** The project relies heavily on `pandas` and `numpy` for data manipulation, alongside `matplotlib`, `seaborn`, and `plotly` for advanced data visualization.
- **Data Footprint:** The dataset consists of 3,000 unique client records across 25 distinct attributes.
- **Data Quality & Cleaning:** An initial inspection confirmed a pristine dataset with exactly zero missing values across all columns. The `Joined Bank` column was successfully parsed from a standard object to a `datetime64[ns]` format to enable time-series analysis.
- **Feature Engineering:** To improve business segmentation, a new feature named `Income Category` was created. `Estimated Income` was bucketed into three distinct tiers:
 - **Low:** < \$100,000
 - **Medium:** \$100,000 to \$300,000
 - **High:** > \$300,000

2. Dashboard Views

To provide stakeholders with an interactive look at the data, the analysis is accompanied by a suite of visual dashboards. Please refer to the following visuals for the complete reporting experience:

- **1. Home Page.jpg:** High-level project overview, KPIs, and main navigation.
- **2. Loan Analysis.jpg:** Detailed breakdown of bank loans, credit card balances, and business lending metrics.

- **3. Deposit Analysis.jpg:** Insights into overall bank deposits, checking/savings account balances, and foreign currency accounts.
- **4. Summary.jpg:** Aggregated portfolio performance and high-level demographic splits.
- **5. Drill Through.jpg:** Deep dive into individual client metrics and historical interactions.

3. Business Intuition: Customer Demographics

Understanding the composition of the bank's clientele is the first step in assessing overall portfolio risk.

- **Income Distribution:** The majority of the client base falls into the **Medium** income tier (1,517 clients), followed by **Low** (1,027), and **High** (456). The overall estimated income spans a wide range, from \$15,919.48 up to a maximum of \$522,330.26.
- **Age Profile:** The average client age is approximately 51 years old. The bank serves a broad demographic, with the youngest client being 17 and the oldest at 85.
- **Geographic & Nationality Split:** European clients represent the largest segment of the portfolio (1,309 clients). This is followed by Asian (754), American (507), Australian (254), and African (176) nationalities.
- **Gender:** The client base is evenly split, comprising 1,512 females (GenderId 2) and 1,488 males (GenderId 1).

4. Business Intuition: Risk & Account Segmentation

By analyzing account types and risk markers, the bank can better tailor its lending and deposit strategies.

- **Risk Weighting:** The portfolio skews toward lower risk. Clients are predominantly assigned a Risk Weighting of 2 (1,222 clients) or 1 (836 clients). Only 160 clients fall into the highest risk category of 5.
- **Loyalty & Fees:** A significant portion of the base (1,476 clients) operates under a 'High' fee structure. In terms of loyalty classifications, 'Jade' is the most common (1,331 clients), followed by 'Silver' (767) and 'Gold' (585). 'Platinum' is the most exclusive tier, held by 317 clients.

- **Asset & Property Ownership:** Real estate exposure is evenly distributed across the client base. 777 clients own 2 properties, 776 own 1 property, 742 own 3 properties, and 705 currently do not own any properties.
- **Credit Utilization:** The majority of clients (1,922) hold only 1 credit card, while 765 clients hold 2, and 313 clients hold 3. The average credit card balance across the portfolio sits at \$3,176.20.